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Analysis of Selected Be-Substituted NiZn Ferrites

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Polycrystalline NiZn ferrites with the chemical formula $(Ni_{0.3}Zn_{0.7})_{1-x}Be_xFe_2O_4$ with Be substituent ion $x = 0, 0.05, 0.1$, and 0.25 have been prepared by the ceramic method. The intrinsic and certain extrinsic magnetic properties such as the coercivity H_c, initial permeability μ_i and hysteresis loops of the ferrites have been measured and discussed from the point of view of the substituting element contents as well as the grain size. The Mössbauer spectra of certain samples have been analysed as well. According to the requirement of specific applications the properties of NiZn, ferrite have to be improved by means of proper substitution of selected ions and processing technology.

Index Terms—**Magnetization, Mössbauer spectra, NiZn ferrite.**

I. Introduction

In the preparation of NiZn ferrites, various kinds of substituents are appropriate to improve resulting properties. The substituent ions can play an important role in determining the intrinsic and extrinsic magnetic properties of NiZn ferrite. The effects of ion substitution can be divided into several types the realization of change distribution of Fe^{3+}, Ni^{2+}, and Zn^{2+} ions in tetrahedral (A) or octahedral (B) sublattice. The amount of these replaced cations is determined by the valence and specific characteristics of the substituting ion. The aim of presented work is the investigation of the effects of NiZn stoichiometry change by partial substitution of both Ni^{2+} and Zn^{2+} cations proportionally with divalent Be cations. These substitutions can influence the microstructure, intrinsic properties, atomic diffusibility and sintering kinetics. This can be interpreted mainly in the view of the replacement of Fe^{3+} ions in B-sites at low concentrations x with Be ions. This is due to the fact that the magnetic behavior of the ferrites is largely governed by Fe-Fe interaction via spin coupling [1]–[3]. The magnetization, Curie temperature and other properties of substituted $(Ni_{1-y}Zn_y)_{1-x}Be_xFe_2O_4$ are also dependent on Zn contents y of initial NiZn ferrite. For various y, the Be cation can have different effect in substituted system at the same x contents. In this paper we would like to discuss the results of the investigation of the effects of $Ni_{0.3}Zn_{0.7}$ ferrite stoichiometry change invoked by partial substitution of both Ni^{2+} and Zn^{2+} with Be^{2+} cations.

II. Experimental Methods

The substituted ferrites with the chemical composition $(Ni_{0.3}Zn_{0.7})_{1-x}Be_xFe_2O_4$ have been synthesized by ceramic way. The samples were sintered at $T_s = 1200, 1250$, and 1300 °C for 6 h in the air and cooled slowly in a furnace. The substituted Be ion concentration was varied as $x = 0, 0.05, 0.1, 0.25$. The temperature dependences of the magnetic susceptibility $\chi(T)$ and the Curie temperatures T_C were measured by the bridge method. The saturation magnetization M_S was measured by means of vibration sample magnetometer. The magnetic phases were verified by Mössbauer spectroscopy and by thermomagnetic analysis of $\chi(T)$. The size and shape of the ferrite powder and grains of sintered samples have been examined by SEM. The hysteresis loops were measured by computer controlled measuring system consisting of commercially available instruments as well as of the hardware and controlling software developed at our institute.

III. Variation of Intrinsic Magnetic Properties

The part of experiments and study was focused on intrinsic properties variation of $(Ni_{0.3}Zn_{0.7})_{1-x}Be_x$ ferrite, i.e., saturation magnetization M_S and Curie temperature T_C. The variation of saturation magnetization M_S as a function of substituent contents x for Be substitution in $(Ni_{0.3}Zn_{0.7})_{1-x}Be_x$ ferrite measured at room temperature is shown in Fig. 1. M_S was found to increase with increasing concentration of substituent x up to a certain value. Similar behavior was published in [4]. Our measured data can be compared qualitatively to well-known behavior of saturation magnetic moments of mixed $Ni_{1-y}Zn_y$ ferrites as a function of zinc concentration y [5], [6], which was measured at 4.2 K. In the mentioned measurements, the saturation magnetization as a function of Zn contents increases with increasing substitution (y), then it goes through the maximum for intermediate values, further decreases with high Zn contents. In our case, due to the substitution with Be, the contents of Zn ions decreases from $y = 0.7$ to $y = 0.7(1 - x)$. As a result, for x within the interval $\langle 0; 0.25\rangle$ the value of saturation magnetization increases. In addition, the Curie temperature T_C has also been found to increase with substituent concentration for Be ions up to $x = 0.25$. The behavior of the T_C as function of x is also shown in Fig. 1. In connection with both curves in Fig. 1 the measured Mössbauer spectra of the samples have been analysed, as well.

One of the views on the analysing of the Mössbauer spectra presented in Fig. 2 is if we suppose that the structure is disordered and we fitted the spectra using two distribution functions. The first corresponds to quadrupole splitting and the second to magnetic field distribution. This model distinguishes nonmagnetic component (N) and magnetic component (M).

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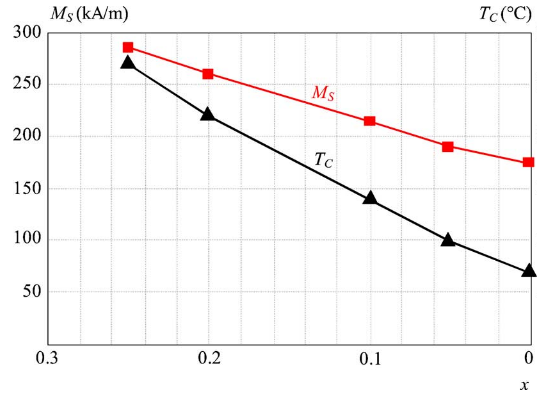

Fig. 1. Saturation magnetization and Curie temperature of substituted NiZn ferrite as a function of Be contents.

The pure $Ni_{0.3}Zn_{0.7}$ ferrite contains nonmagnetic components with average quadrupole splitting $QS = 0.49$ mm/s and relative spectral areas $A_N = 33\%$ and magnetic components with average internal magnetic field $B_{\text{eff}} = 14.7$ T and relative spectral areas $(A_A + A_B)$ of 67%. The addition of Be^{2+} ions influences all parameters. Sample with the concentration 0.05 of Be [Fig. 2(a)] shows increasing of average quadrupole splitting and value of average magnetic field at $QS = 0.56$ mm/s and $B_{\text{eff}} = 19.7$ T. The relative spectra area changed at $A_N = 20\%$ and $A_M = 80\%$, respectively. By further addition of 0.25 Be [Fig. 2(b)] nonmagnetic components rapidly decreased at $A_N = 2\%$. The magnetic components increased at $A_M = 98\%$. The same tendency had also quadrupole splitting $QS = 0.16$ mm/s and internal magnetic field $B_{\text{eff}} = 29.2$ T. These changes indicate that the substitution of Be^{2+} ions have an influence on the super exchange interactions between Aand B sublattices.

In Fig. 2, the Mössbauer spectra show that the substitution of Be^{2+} ions increases the superexchange interactions between Aand B sublattices. Due to the fact, that the surroundings of Fe^{3+} ions in B sublattices have changed, the magnetic structure has changed as well. The presence of Be^{2+} ions has the positive influence on magnetic interactions between Aand B sublattices. If a part of Zn and Ni ions is substituted with Be^{2+} ions, it has to be taken into account that for $x \leq 0.25$ Be ions can probably occupy the octahedral (B) lattice sites. We have assumed then that the incorporation of Be ions into the basic composition $(Ni_{0.3}Zn_{0.7})_{1-x}Be_x$ yields the cation distribution description according to schema

$$\left(Zn^{2+}_{0.7(1-x)}Fe^{3+}_{0.3+0.7x}\right)\left[Ni^{2+}_{0.3(1-x)}Be^{2+}_{x}Fe_{1.7-0.7x}\right]O_4^{2-} \quad (1)$$

and the theoretical magnetic moment can be calculated as

$$\overline{(0+5(0.3+0.7x))} >< \overline{[2.2(0.3(1-x))+0+5(1.7-0.7x)]} \quad (2)$$

where taking Fe^{3+} magnetic moments as 5 μ_B and 2.2 μ_B is the moment of Ni in Bohr magnetons. This theoretical schema indicates that the value the net magnetic moment for $x = 0$ reaches the value of 7.66 μ_B.

The measured saturation magnetic moment (at temperature 4.2 K) had lower value of 4.6 μ_B due to B-site canting into two sublatices B′ and B″ [7], i.e., Yaffett-Kittel splitting. The reason is that the nonmagnetic Zn ions substitute the magnetic Fe ions resulting in weakening the A-B interaction due to high magnetic dilution of tetrahedral (A) lattice. The Be substitution x provides that the concentration of Ni^{2+} ions decreased by the adequate of Be^{2+} ions in B-positions and the migration of (0.7 x)Fe^{3+} ions from B-sites to A-sites is provided by residual contents of Be^{2+} ions. The strength of A-B interaction was increased with x since it depends on the increase of Fe ions in A-positions. Thus, A-sites are able to align more moments of B-lattice ions antiparallel one to another, which increases the magnetization of B-lattice due to decrease of canting angle between magnetization vectors B′ and B″ sublattices. Therefore, total magnetization increases with each substitution step up to critical value x_c and then slightly decreases. In agreement with this idea the measured values of M_S increasing with $x = 0$ up to $x = 0.25$ is presented in Fig. 1 (at temperature 300 and 4, 2 K). In addition, the Curie temperature increases with strengthening A-B interaction, i.e., increasing with x (within investigated substitution interval), see Fig. 1. This results in the increase of thermal stability of Be-doped NiZn ferrite. The measurements of temperature dependencies of magnetic susceptibility $\chi(T)$ confirmed the increase of Curie temperature value from 70 °C at pure $Ni_{0.3}Zn_{0.7}$ ferrite without Be^{2+} up to 268 °C for $x = 0.25$. Similar behavior of T_C versus x was observed by [4] for $x \in \langle 0; 0.25\rangle$. If $x > 0.25$ the T_C versus x is decreasing, it is in agreement with [8], where from X-ray diffraction analysis was concluded

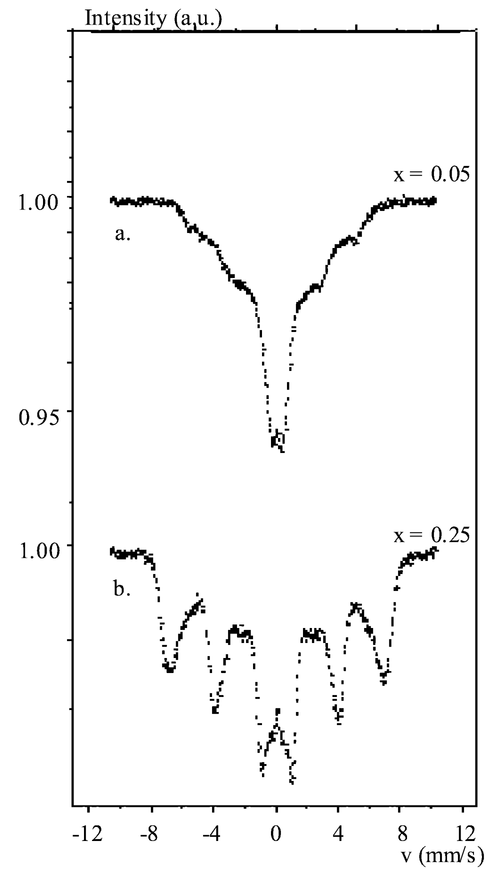

Fig. 2. Mössbauer spectra of $(Ni_{0.3}Zn_{0.7})_{1-x}Be_xFe_2O_4$ ferrites with $x = 0.05$ (a) and 0.25 (b).

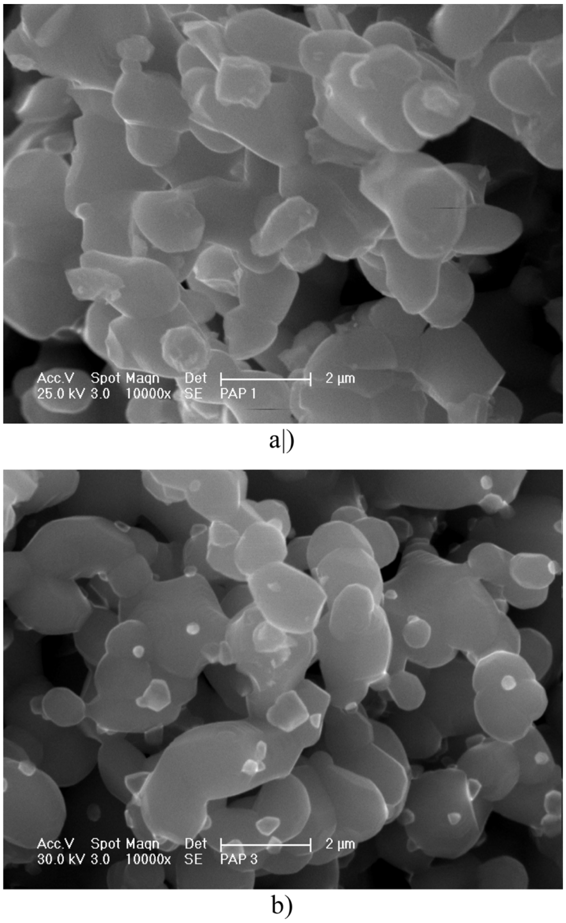

Fig. 3. Examples of SEM images of $(\mathrm{Ni}_{0.3}\mathrm{Zn}_{0.7})_{1-x}\mathrm{Be}_x\mathrm{Fe}_2\mathrm{O}_4$ ferrites with $x = 0.05$ (a) and 0.1 (b).

that the haematite is created along with the spinel phase for $x > 0.25$. This opinion can be approved by our T_{C} measurements. In addition, the Mössbauer measurements did not give an evidence for the presence of the haematite in our case for the substitution up to $x = 0.3$.

IV. Modification of Extrinsic Properties Owing to Be Substitution

Another part of experiments aimed at the examination of the changes of coercivity, permeability and hysteresis loops with Be substitution contents x of $(\mathrm{Ni}_{0.3}\mathrm{Zn}_{0.7})_{1-x}\mathrm{Be}_x$ ferrite. It is known that the shape, surface, roughness, cracks, pores, impurities, etc. play important role in ferrites grain size. Samples properties influence the demagnetizing field anisotropy due to these items. Selected properties as coercivity H_c, permeability μ_{i} and average grain size $\langle D \rangle$ as dependence of x, Be contents for the samples sintered at 1200 °C are presented in Table I. Examples of SEM images are in Fig. 3.

By increasing Be content the grain size and permeability decrease and on the other hand the coercivity increases. Therefore, it can be assumed, that internal magnetic field is increasing due to the growth of the demagnetizing field.

Strong influence of Be substitution amount and sintering temperature on the hysteresis loops is shown in Figs. 4 and 5 where the hysteresis loops are measured at the frequency $f = 100$ Hz and maximum exciting field of 25 and 1000 A/m, respectively. The shape as well as the basic parameters of the loops (such as the coercivity, maximum and/or remanent flux density) are significantly dependent on Be concentration and sintering temperature. One can see that regardless of the maximum field applied the loops of materials sintered at 1200 °C are different from those of materials with the same chemical composition sintered at higher temperatures (1250 °C and 1300 °C). Thus, the sintering temperature of 1250 °C seems to be enough to reach final magnetic properties. The effect of the sintering temperature on the hysteresis loop will be the subject for further studies in the laboratory. The above mentioned influence of demagnetizing field associated with particle size and/or the porosity is clearly visible in weak fields (Fig. 4) where the loops of the samples with $x = 0.1$ and 0.25 exhibit almost no hysteresis, even if the saturated coercivity is directly proportional to the distribution of grain sizes as well as x.

TABLE I
Selected Properties of NiZnBe Ferrite

x (-)	H_{c} (A/m)	μ_{i} (-)	<D> (μm)
0	25	2750	15
0.05	55	800	5
0.1	280	370	3
0.25	500	110	2

Note that the value of maximum flux density $B_{\max}$ at given maximum applied field $H_{\max} = 1000$ A/m (Fig. 5) approaches the largest values for $x = 0.1$ which is in contradiction with the dependence of saturation magnetization M_S upon x shown in Fig. 1 (nearly linear increase with x). This behavior is similar to that observed in case of Cu-substituted ferrites with the same chemical formula (except for Cu instead of Be) [9]. The differences can be explained by the fact that the saturation magnetization is an intrinsic (thus, structurally independent) parameter meanwhile the maximum flux density corresponds to structurally sensitive approaching the saturation.

V. Conclusion

The influence of Be substitution on the intrinsic magnetic properties of $(\mathrm{Ni}_{0.3}\mathrm{Zn}_{0.7})_{1-x}\mathrm{Be}_x\mathrm{Fe}_2\mathrm{O}_4$, i.e., saturation magnetization and the Curie temperature as well as some extrinsic magnetic properties, i.e., coercivity and initial permeability have been analysed. The experiments have shown strong effect of Be substitution and sintering temperature on the resulting parameters of sintered ferrites thanks to close relationship between the chemical composition and resulting crystalline structure and grain-size distribution of the samples. The Curie temperature is primarily determined by the strongest interaction, the interaction between A-B sublattices being dominant in this case. The Curie temperature, therefore, increases with increasing number of A-B interactions, i.e., with increasing Be contents x.

The saturation magnetization increases with x up to certain critical value. The Mössbauer spectra confirmed the effect associated with increasing the Curie temperature and saturation magnetization due to decreasing the Zn contents (which is

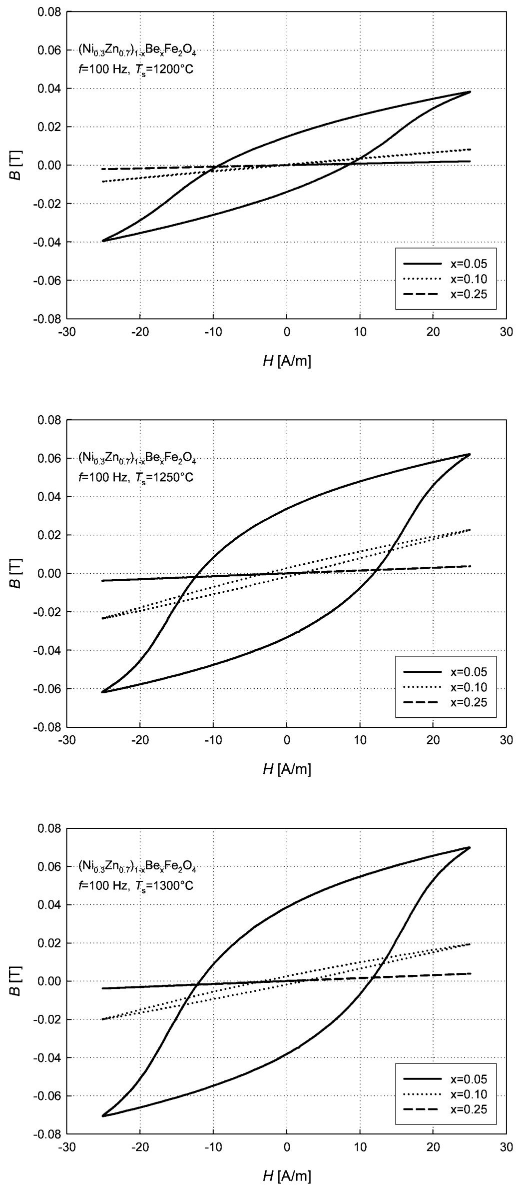

Fig. 4. Hysteresis loops of $(\mathrm{Ni}_{0.3}\mathrm{Zn}_{0.7})_{1-x}\mathrm{Be}_x\mathrm{Fe}_2\mathrm{O}_4$ ferrites with $x = 0.05, 0.1, 0.25$, sintering temperature $T_s = 1200, 1250$, and 1300 °C, $H_{\max} = 25$ A/m.

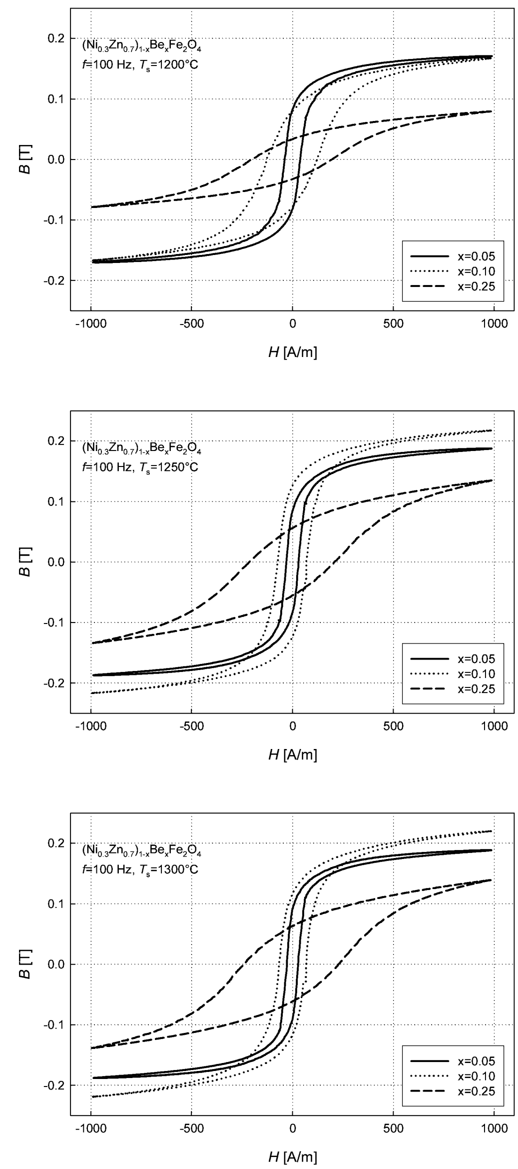

Fig. 5. Hysteresis loops of $(\mathrm{Ni}_{0.3}\mathrm{Zn}_{0.7})_{1-x}\mathrm{Be}_x\mathrm{Fe}_2\mathrm{O}_4$ ferrites with $x = 0.05, 0.1, 0.25$, sintering temperature $T_s = 1200, 1250$, and 1300 °C, $H_{\max} = 1000$ A/m.

replaced by Be). In accordance with saturation magnetization measurements, the maximum flux density of hysteresis loops measured at relatively high exciting field amplitudes exhibits a maximum at certain value of x_c. From the practical point of view, a novel type of NiZnBe ferrite has been prepared with appropriate B_s and high initial low-frequency permeability with only a small increase of the coercivity. As we have already shown in [4], by proper choice of the relation between the grain size, particle shape and Be contents, a NiZnBe ferrite with required properties for microwave applications can be prepared. In addition, an appropriate NiZnBe ferrite can be used as the filler into a polymer matrix to produce a composite material with the permeability dispersion peak shifted to the frequency range of a few GHz.

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